

COURSE PLAN

(Undergraduate Programme)

(Each Teacher to submit course plan individually)



F/QSP 11/10/06

DEPARTMENT: Computer Science & Information Technology

Course Code: CT-175

Course Title: Programming Fundamentals Teacher: Furqan Hussain

Semester: Spring/Fall

Academic Session: 2025

Class: FSCT (Section A)

No. of sections: 2

No of contact hours per week per section: 3

TEACHING TOOLS

☒ Problem-based learning	☒ Mini-project	☐ Case studies
☐ Innovative solution	☒ Group activity	☐ Entrepreneurial activities
☐ Industrial visit	☐ Student presentation	☐ Other _____

Week No.	TOPICS	NO. OF PERIODS
1.	Introduction to Programming, Pseudo-codes, Flowchart, C Language	03
2.	Variables and their Scope, Constants, Data types, Format Specifier, Keywords, Simple Program, Linking, Compiling, and Loading	03
3.	Operators – Arithmetic, Relational, Logical, Bit-Wise, Increment, Decrement Operators	03
4.	Conditional Structure – if-else Construct, Nested if-else, Ladder if-else-if, Ternary operator	03
5.	Conditional Structure – switch Construct, Nested Switch, break Statement	03
6.	Iterative Structure – for Construct, while Construct, do-while Construct, break and continue Statements	03
7.	Nested Iterative Structure, Pattern Printing	03
8.	Mid – Term Examinations	03
9.	Array - 1D and 2D Arrays, Character Array	03
10.	Strings – String Processing using Built in Library Functions	03
11.	Function – Declaration, Definition, Function Call, Passing Arguments by Value and by Reference, static Variable	03
12.	Pointers – Introduction, void & constant Pointers, Pointers and Array	03
13.	Pointers – Passing to and Returning from Function, Dynamic Memory Allocation using calloc(), malloc(), free()	03
14.	Structures – Simple Structure, Nested Structure, Passing to and Returning from Function	03
15.	File Handling – Create, Open, Edit, Read and Write on Text file	03
Total per Semester:		45

SESSIONAL CRITERIA

Assessment Type	Marks	Schedule (Week No.)
Mid Term Exam	20	8 th
Quizzes	05	4 th , 12 th
Home Assignment/Code In Place/Leet Code	05 + 05	10 th
CCP / Project Report	05	14 th
Total Sessional Marks	40	

Prepared by: Furqan Hussain Essani

Reviewed & Approved by: Prof. Dr. M. Mubashir Khan (CCS)

Date _____

Date _____